

Universal dimension sequences of pairwise Poisson algebras: independence from spatial dimension, potential exponent, and charge sign

Brian J. Sheppard

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Abstract

We extend the pairwise Poisson algebra framework introduced in [1] from the planar three-body problem to N bodies in d spatial dimensions, with arbitrary singular potentials and charge configurations. The four-body ($N=4$) dimension sequence

$$d_4(0) = 6, \quad d_4(1) = 14, \quad d_4(2) = 62$$

is computed for the first time, with an SVD gap ratio of 3.4×10^{11} at level 2. This sequence is mass-invariant (verified at three mass configurations spanning all symmetry types) and independent of the spatial dimension d (verified at $d = 1, 2, 3$).

For $N=3$, the known sequence [3, 6, 17, 116] is shown to be independent of the pole order of the potential: the $1/r$, $1/r^2$, and $1/r^3$ potentials all yield the identical sequence. Charge-sign invariance is established: all-attractive, all-repulsive, and mixed helium-like ($q = +2, -1, -1$) charge configurations produce the same dimension sequence.

These results support the **Universality Conjecture**: the dimension sequence of the pairwise Poisson algebra depends only on N and whether the potential's exponent lies in the exceptional set $\{r^1, r^2, r^3\}$, rather than on a singular-versus-regular dichotomy, and is independent of the spatial dimension, mass ratios, pole order, and the sign of the interaction. Regularity alone does not imply finiteness: among integer-power polynomial potentials, only r^1 , r^2 , and r^3 are exceptional (closing at dimension 5 and 15 respectively, or departing non-nilpotently at dimension 109), while r^4 through r^{10} reproduce the singular sequence [3, 6, 17, 116]; the harmonic (r^2) closure reflects an enhanced dynamical symmetry, not regularity *per se*. Evidence is presented at two values of N ($N=3$ and $N=4$), three spatial dimensions ($d=1, 2, 3$), three singular potentials ($1/r$, $1/r^2$, $1/r^3$), and three charge configurations.

1 Introduction

In [1] we introduced the pairwise Poisson algebra $\mathcal{A}$ generated by the three interaction Hamiltonians H_{12}, H_{13}, H_{23} of the planar three-body problem under iterated Poisson brackets. The cumulative dimension sequence

$$d_3(0) = 3, \quad d_3(1) = 6, \quad d_3(2) = 17, \quad d_3(3) = 116$$

grows super-exponentially, consistent with conjectured infinite Gelfand–Kirillov dimension. The sequence is mass-invariant, and identical for the Newtonian ($1/r$) and Calogero–Moser ($1/r^2$) potentials, while the integrable harmonic potential (r^2) produces a finite-dimensional algebra stabilising at dimension 15. In [2] we revealed the internal organisation of this algebra: a four-tier decomposition $52 + 44 + 16 + 4 = 116$ explained by S_3 representation theory and a jet filtration with integer-quantized scaling exponents.

The present paper establishes five new independence results that together motivate a precise universality conjecture:

1. **N -body extension.** The $N=4$ dimension sequence [6, 14, 62] is computed for the first time (Section 3).
2. **Mass invariance at $N=4$.** The sequence is verified at three mass configurations spanning all symmetry types (Section 4).
3. **Spatial dimension independence.** For both $N=3$ and $N=4$, the sequences are identical at $d = 1, 2, 3$ (Section 5).
4. **Potential exponent independence.** The $1/r$, $1/r^2$, and $1/r^3$ potentials all produce the sequence [3, 6, 17, 116] for $N=3$ (Section 6).
5. **Charge-sign invariance.** Flipping interaction signs (attractive $\leftrightarrow$ repulsive) does not change the dimension sequence (Section 7).

In Section 8 we formulate the Universality Conjecture: the dimension sequence depends only on N and whether the potential's exponent lies in the exceptional set $\{r^1, r^2, r^3\}$, and nothing else.

2 The N -body engine

2.1 Generalisation to N bodies

Consider N point masses in d spatial dimensions with phase space $\mathbb{R}^{2Nd}$ and canonical coordinates (q_i^α, p_i^α) , $i = 1, \dots, N$, $\alpha = 1, \dots, d$. The $\binom{N}{2}$ pairwise interaction Hamiltonians are

$$H_{ij} = T_i + T_j + V(r_{ij}), \quad T_i = \sum_{\alpha=1}^d \frac{(p_i^\alpha)^2}{2m_i},$$

where $r_{ij} = (\sum_{\alpha} (q_i^\alpha - q_j^\alpha)^2)^{1/2}$ and $V(r) = -c_{ij} r^{-p}$ for some power $p > 0$ and coupling constants c_{ij} . The polynomial representation $u_{ij} = 1/r_{ij}$ renders all expressions polynomial in the extended variables $(q_i^\alpha, p_i^\alpha, u_{ij})$, with chain rule

$$\frac{\partial u_{ij}}{\partial q_i^\alpha} = -(q_i^\alpha - q_j^\alpha) u_{ij}^3.$$

This identity is independent of d : the same algebraic structure governs brackets in one, two, and three spatial dimensions.

Definition 1. *The N -body pairwise Poisson algebra $\mathcal{A}^{(N,d)}$ is the Lie algebra generated by the $\binom{N}{2}$ pairwise Hamiltonians under the Poisson bracket, with the filtration*

$$\begin{aligned} \mathcal{A}_0^{(N,d)} &= \text{span}\{H_{ij} : 1 \leq i < j \leq N\}, \\ \mathcal{A}_n^{(N,d)} &= \mathcal{A}_{n-1}^{(N,d)} + \text{span}\{\{f, g\} : f \in \mathcal{A}_{n-1}^{(N,d)}, g \in \mathcal{A}_{n-1}^{(N,d)}\}, \end{aligned}$$

and dimension function $d_N(n) = \dim \mathcal{A}_n^{(N,d)}$.

2.2 Computational method

All brackets through the target level are computed symbolically using exact rational arithmetic in SymPy. At each level, the accumulated generators are evaluated numerically at randomly sampled phase-space points, and the rank is determined by SVD gap analysis. A gap ratio $\sigma_k/\sigma_{k+1} > 10^4$ is required for a definitive rank determination.

The engine is validated by reproducing the known $N=3$ sequence [3, 6, 17, 116] with SVD gaps exceeding 10^9 at every level (above 10^{10} at levels 0–2; the level-3 gap tops out at $\approx 5 \times 10^9$). The dimension values themselves are exact ranks over $\mathbb{Q}$; the SVD is a cross-check, not the proof.

3 The $N=4$ dimension sequence

Theorem 2 ($N=4$ dimensions). *For four bodies in the plane ($N=4, d=2$) with the Newtonian potential $V = -1/r$:*

Level	$d_4(n)$	New	Candidates	Gap ratio	Status
0	6	6	6	—	exact
1	14	8	15	$> 10^{10}$	exact
2	62	48	—	3.4×10^{11}	exact

Proof (computational). Level 0 has $\binom{4}{2} = 6$ pairwise Hamiltonians. At level 1, there are $\binom{6}{2} = 15$ candidate brackets $\{H_{ij}, H_{kl}\}$. Three of these vanish identically: the pairs with disjoint body indices $\{H_{12}, H_{34}\}$, $\{H_{13}, H_{24}\}$, and $\{H_{14}, H_{23}\}$ have zero Poisson bracket because the constituent Hamiltonians share no canonical variables. Of the remaining 12 nonzero brackets, one additional linear dependence exists, yielding $d_4(1) = 14$. At level 2, SVD analysis with 500 phase-space samples gives rank 62 with a gap ratio of 3.4×10^{11} , establishing $d_4(2) = 62$. $\square$

Remark 3 (Growth comparison). The growth ratios for $N=4$ exceed those for $N=3$ at every level:

Ratio	$N=3$	$N=4$
$d(1)/d(0)$	2.00	2.33
$d(2)/d(1)$	2.83	4.43

The acceleration of growth with N is consistent with the larger interaction graph (K_4 has 6 edges vs. K_3 's 3) providing more algebraically independent bracket combinations.

Remark 4 (Disjoint-pair vanishing). The vanishing of $\{H_{12}, H_{34}\}$ is a consequence of the pairwise structure: H_{12} depends on bodies 1 and 2 only, while H_{34} depends on bodies 3 and 4 only. Since they share no canonical variables, their Poisson bracket is identically zero. This phenomenon is absent for $N=3$ (where every pair of Hamiltonians shares at least one body) and becomes increasingly prevalent for larger N : K_N has $\binom{N}{2}$ edges, of which $\binom{N}{2}\binom{N-2}{2}/2$ disjoint pairs bracket to zero.

4 Mass invariance at $N=4$

Theorem 5 (Mass invariance, $N=4$). *The dimension sequence [6, 14, 62] is mass-invariant for $N=4$. Verification:*

Configuration	(m_1, m_2, m_3, m_4)	$d_4(0:2)$
Equal (S_4)	(1, 1, 1, 1)	6, 14, 62
Hierarchical	(100, 10, 1, 1)	6, 14, 62
Mixed (no symmetry)	(3, 7, 11, 2)	6, 14, 62

The proof follows the same Zariski argument as for $N=3$ [1]: the generators are polynomials in $(q_i^\alpha, p_i^\alpha, u_{ij})$ whose coefficients are rational functions of the masses. The rank of the evaluation matrix is constant on a Zariski-open dense subset of the positive mass cone, and can only drop on a proper algebraic subvariety. The three tested configurations — equal masses (maximal S_4 symmetry), hierarchical (broken symmetry, extreme ratio 100:1), and fully generic — span all symmetry types without exhibiting any rank drop.

5 Spatial dimension independence

Theorem 6 (*d*-independence). *For both $N=3$ and $N=4$, the dimension sequence is independent of the spatial dimension d :*

N	d	Phase space	Sequence
3	1	$\mathbb{R}^6 + 3 \text{ aux}$	3, 6, 17, 116
	2	$\mathbb{R}^{12} + 3 \text{ aux}$	3, 6, 17, 116
	3	$\mathbb{R}^{18} + 3 \text{ aux}$	3, 6, 17, 116
4	1	$\mathbb{R}^8 + 6 \text{ aux}$	6, 14, 62
	2	$\mathbb{R}^{16} + 6 \text{ aux}$	6, 14, 62
	3	$\mathbb{R}^{24} + 6 \text{ aux}$	6, 14, 62

All gap ratios exceed 10^9 , with ratios above 10^{10} at levels 0–2 (the $N=3$ level-3 gap is bounded as in Section 2).

This result was unexpected. The original conjecture (stated informally in [1]) predicted that the sequence should depend on d , since higher spatial dimensions provide more degrees of freedom and a larger function space. The $d=1$ case is particularly surprising: with only one spatial coordinate per body, the phase space $\mathbb{R}^{2N}$ is much smaller than the $d=3$ phase space $\mathbb{R}^{6N}$, yet the algebra achieves the same dimension.

Remark 7 (Interaction graph interpretation). The d -independence suggests that the algebra detects the *combinatorial topology of the interaction graph* K_N rather than the ambient geometry. The complete graph on N vertices has $\binom{N}{2}$ edges, each carrying a pairwise Hamiltonian. The bracket structure — which pairs of edges share a vertex — determines which brackets are nonzero. This combinatorial data is independent of d .

The polynomial representation reinforces this interpretation: the chain rule $\partial u_{ij}/\partial q_i^\alpha = -(q_i^\alpha - q_j^\alpha) u_{ij}^3$ has the same algebraic form regardless of d . Increasing d adds more terms to the Poisson bracket sum but does not change which u_{ij} monomials appear in the generators, and it is the monomial structure that determines linear independence.

6 Potential exponent independence

Theorem 8 (Pole-order invariance). *For $N=3$, $d=2$, the potentials $V = -1/r^p$ with $p = 1, 2, 3$ all produce the identical dimension sequence:*

Potential	$d(0)$	$d(1)$	$d(2)$	$d(3)$
$1/r$ (Newton)	3	6	17	116
$1/r^2$ (Calogero–Moser)	3	6	17	116
$1/r^3$	3	6	17	116
r^2 (harmonic)	3	6	13	15

The harmonic potential is included for contrast: its algebra stabilises at dimension 15 with zero new generators at level 4 [1].

Remark 9 (The singular/regular dichotomy). In the polynomial representation, the potential $V_{ij} = -c_{ij} u_{ij}^p$ is a monomial of degree p in the auxiliary variable u_{ij} . The chain rule derivative $\partial u_{ij}/\partial q_i^\alpha = -(q_i^\alpha - q_j^\alpha) u_{ij}^3$ introduces u_{ij}^3 , regardless of p . Iterated brackets generate expressions with increasing powers of u_{ij} , and the exponent p of the potential merely shifts the initial power.

For the harmonic potential $V_{ij} = c_{ij} r_{ij}^2$, the situation is qualitatively different: V_{ij} is polynomial in the coordinates (no auxiliary variable needed), and all iterated brackets remain polynomial. The algebra closes not because polynomial rings are finitely generated at each degree —

r^4 through r^{10} are equally polynomial and do not close — but because the harmonic oscillator possesses an enhanced dynamical symmetry (a Fradkin-tensor / $SU(d)$ structure) special to the degree-2 case.

The relevant structural distinction is therefore:

- **Singular potentials** ($V \propto r^{-p}$, $p > 0$): infinite-dimensional algebra, identical dimension sequence regardless of p .
- **Regular (polynomial) potentials** (V polynomial in r): finiteness is the exception, not the rule. Only r^1 , r^2 , and r^3 are exceptional (closing at dimension 5 and 15 respectively, or departing non-nilpotently at dimension 109); r^4 through r^{10} instead reproduce the singular sequence [3, 6, 17, 116]. The harmonic (r^2) closure is attributable to an enhanced dynamical symmetry, not to regularity *per se*.

7 Charge-sign invariance

Theorem 10 (Charge-sign invariance). *For $N=3$, $d=3$, $V = 1/r$, the dimension sequence is independent of the sign and magnitude of the pairwise coupling constants:*

Configuration	Charges (q_1, q_2, q_3)	Sequence
All-attractive (gravity)	—	3, 6, 17, 116
Full helium	(+2, −1, −1)	3, 6, 17, 116
All-repulsive	(+1, +1, +1)	3, 6, 17, 116

In the helium configuration, H_{12} and H_{13} (nucleus–electron) are attractive while H_{23} (electron–electron) is repulsive.

Proof sketch. The coupling constant $c_{ij} = q_i q_j$ enters the potential as $V_{ij} = c_{ij} u_{ij}^p$. In the Poisson bracket, the coupling appears as a multiplicative factor on terms involving u_{ij} :

$$\{H_{ij}, H_{kl}\} = c_{ij} c_{kl} \cdot (\text{bracket of monomial parts}).$$

The bracket of the monomial parts depends on *which* variables appear (which edges of K_N share a vertex) but not on the numerical values of the couplings. Two generators are linearly dependent if and only if the monomial parts are dependent, and this condition is independent of the c_{ij} . The rank of the evaluation matrix is therefore invariant under arbitrary nonzero rescaling of the couplings, including sign changes. $\square$

Remark 11 (Metric vs. topological invariance). While the *dimension* (rank) is charge-invariant, the *SVD gap ratio landscape* over configuration space does depend on charges. For $N=3$ with the $1/r$ potential, the Pearson correlation between the charged and uncharged gap maps is $r \approx 0.85$. For $1/r^2$, the correlation drops to $r \approx 0.76$. This is a *metric* distinction (how cleanly the $d_3(3) = 116$ -dimensional subspace separates from the null space) rather than a *topological* one (which subspace it is).

The stronger charge sensitivity of $1/r^2$ is consistent with the quadratic coupling: charges multiply u_{ij}^2 rather than u_{ij} , producing more complex cross-derivative interactions.

8 The universality conjecture

The five independence results of this paper, combined with the mass invariance and potential comparison of [1], motivate the following:

Conjecture 12 (Universality). *For $N \geq 3$ particles in $d \geq 1$ spatial dimensions, interacting via a singular central potential $V(r) = c r^{-p}$ with $p > 0$ and arbitrary nonzero coupling constants c_{ij} :*

1. The pairwise Poisson algebra $\mathcal{A}^{(N,d)}$ has infinite Gelfand–Kirillov dimension.
2. The cumulative dimension sequence $d_N(n)$ depends only on N , not on d , the masses m_i , the pole order p , or the couplings c_{ij} .

For regular potentials (V polynomial in r), finiteness is not generic: only r^1 and r^2 close to a finite-dimensional algebra (at dimension 5 and 15 respectively). The exponent r^3 is also exceptional — it departs from the universal sequence at dimension 109 — but does not close; r^4 through r^{10} are not exceptional and reproduce the universal sequence.

8.1 Evidence summary

Table 1 collects all computational evidence for the conjecture.

Parameter varied	N	Range tested	Result
Masses	3	4 configs incl. Tsygvintsev	invariant [1]
Masses	4	3 configs (equal, hierarchical, mixed)	invariant
Spatial dim. d	3	$d = 1, 2, 3$	invariant
Spatial dim. d	4	$d = 1, 2, 3$	invariant
Pole order p	3	$p = 1, 2, 3$	invariant
Charge signs	3	attractive, repulsive, mixed	invariant
Exceptional exponents	3	r^1, r^2, r^3 vs. all others	departs vs. universal
N	3, 4	—	different sequences

Table 1: Evidence for the Universality Conjecture. The dimension sequence changes only with N or at the exceptional exponents r^1, r^2, r^3 . All other parameters leave it invariant.

8.2 What the conjecture does not claim

The conjecture is a statement about the *algebraic structure* of the pairwise Hamiltonian decomposition, not about the dynamics. In particular:

- It is **not** a non-integrability certificate. As shown in [1], the completely integrable Calogero–Moser system ($1/r^2$, one dimension) produces the same dimension sequence as the non-integrable gravitational three-body problem.
- The dimension sequence does not distinguish integrable from non-integrable systems within the singular class. Whether finer algebraic data (SVD gap magnitudes, structure constants, tier decomposition) can make this distinction remains open.

8.3 The interaction graph interpretation

The conjecture admits a concise reformulation in graph-theoretic language. Associate to the N -body system the *interaction graph* K_N (complete graph on N vertices), with each edge carrying a pairwise Hamiltonian. The Poisson bracket structure depends only on the *edge-adjacency* of K_N : two Hamiltonians H_{ij} and $H_{k\ell}$ bracket nontrivially if and only if $\{i, j\} \cap \{k, \ell\} \neq \emptyset$ (their edges share a vertex).

Remark 13. In the graph-theoretic language, the conjecture states that the dimension sequence $d_N(n)$ is a *graph invariant* of K_N equipped with an exceptional-exponent label (whether the potential is one of r^1, r^2, r^3 , or lies outside this set). All other data — the embedding dimension d , the vertex weights (masses), the edge weights (couplings and pole order) — are irrelevant.

8.4 Falsifiable predictions

The conjecture makes concrete, testable predictions:

1. $N=5$, $d=2$, $1/r$: $d_5(0) = \binom{5}{2} = 10$. The conjecture predicts that $d_5(1)$ and $d_5(2)$ are mass-invariant. Computing these values would extend the evidence to a third value of N .
2. $N=4$, $d=2$, $1/r^2$: *Resolved through level 2*. The Calogero–Moser potential reproduces $[6, 14, 62]$, matching $1/r$; the exact level-3 value $d_4(3) = 1,260$ has been established for $1/r$ only.
3. $N=3$, **level 4**: does $d_3(4)$ agree for $1/r$ and $1/r^2$? The lower bound $d_3(4) \geq 5,604$ (200,000 samples, not definitive) [1] was established for $1/r$ only. An independent computation for $1/r^2$ at level 4 would test whether the sequences remain identical at deeper bracket levels.

9 Discussion

9.1 Relationship to Papers 1 and 2

This paper completes a trilogy. Paper 1 [1] established the dimension sequence $[3, 6, 17, 116]$ and its mass invariance for $N=3$. Paper 2 [2] revealed the internal organisation of the 116-dimensional algebra via S_3 representation theory and jet filtration. The present paper shows that the sequence is far more robust than initially expected: it is independent of every tested parameter except N and whether the potential’s exponent lies in the exceptional set $\{r^1, r^2, r^3\}$.

9.2 S_N structure for $N=4$

Paper 2 showed that the S_3 isotypic decomposition explains the tier structure $52 + 44 + 16 + 4 = 116$ for $N=3$. For $N=4$, the relevant symmetry group is S_4 , which has five irreducible representations of dimensions $1, 1, 2, 3, 3$ (corresponding to the partitions of 4). Whether the 62-dimensional level-2 algebra for $N=4$ exhibits an analogous tier decomposition governed by S_4 Clebsch–Gordan rules is a natural open question.

9.3 Cosmological perspective

If the conjecture holds for general N , it has a suggestive consequence for gravitational N -body systems at cosmological scales.

The symmetric group S_N acts on the reduced configuration space of N bodies. The maximally symmetric configuration — all bodies distributed uniformly — is a fixed point of this action. The critical locus conjecture of [1] predicts a rank drop at S_N -symmetric configurations.

For the universe ($N \sim 10^{80}$ baryonic particles interacting via $1/r$ gravity), the homogeneous, isotropic distribution is such a fixed point. The dramatic dimensional reduction from $\sim 3N$ degrees of freedom to a single scale factor $a(t)$ governed by the Friedmann equation is consistent with an algebraic rank drop at the S_N -symmetric stratum.

We emphasise the caveats: general relativity replaces Newtonian gravity at cosmological scales, the conjecture is unproved for $N > 4$, and the identification of surviving generators with specific cosmological observables is an open problem. The cosmological perspective is offered as motivation for extending the conjecture to large N , not as a claim about the physical universe.

9.4 Open questions

1. **$N=5$ and beyond.** *Partially resolved*. The dimension sequence has been computed through level 2: $d_5(0) = 10$, $d_5(1) = 25$, $d_5(2) = 145$, extending the evidence to a third value of N (mass invariance at $N=5$ has not been separately verified across multiple configurations; see the falsifiable prediction above). Level 3 for $N=5$ remains an open computational target.
2. $d_4(3)$. *Resolved for $1/r$* . $d_4(3) = 1,260$ exactly, giving growth ratio $d_4(3)/d_4(2) \approx 20.3$, exceeding the acceleration seen at $N=3$ ($d_3(3)/d_3(2) = 6.82$). Whether other potentials

(e.g. $1/r^2$) match this exact value at level 3 remains untested (verified only through level 2; see above).

3. **Higher-level divergence.** The sequences for different pole orders match through level 3 for $N=3$. Could they diverge at level 4 or beyond? The conjecture predicts they do not, but this remains unverified.
4. **Logarithmic and Yukawa potentials.** *Resolved.* The $1/r^p$ family does not exhaust all singular potentials, but logarithmic ($V \propto \log r$, relevant for 2D gravity and vortex dynamics) and Yukawa ($V \propto e^{-\alpha r}/r$, relevant for screened Coulomb interactions) potentials have since been tested and both reproduce the universal sequence [3, 6, 17, 116].
5. **Non-complete interaction graphs.** All results in this paper concern the complete graph K_N . What happens if only a subset of pairwise interactions is present (e.g., nearest-neighbour chains)? The conjecture as stated applies only to K_N ; extensions to other graphs would require new computations.

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